



Optofluidic refractive index sensor based on asymmetric diffraction

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Abstract: A novel optofluidic refractive index (RI) sensor was proposed based on asymmetric Fraunhofer diffraction. In-plane optofluidic lens, light source, slit, diffraction pattern visualization zone and optical path were integrated into the microfluidic networks to avoid the manual alignment of the optical components as well as to reduce the cost of external bulky components. Unlike the conventional RI sensor, this device visualizes the bulk refractive index change of the liquid through a diffraction image, which is readily read-out for clinical diagnosis right at the point-of-care or on-site security check. In the experiment, the device can measure a RI change of as low as $\sim 10^{-5}$ RIU. A low noise-equivalent detection limit (NEDL) of $\sim 10^{-6}$ refractive index unit (RIU) and high sensitivity of $\sim 1.1 \times 10^4$ /RIU were achieved. The new device is practical and suitable to be extended for high throughput applications by simultaneously reading multiple chips with an 2D-array image sensor.

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1. Introduction

Optofluidic refractive index sensors are extensively investigated for the applications in medical diagnosis, environmental monitoring and homeland security [1,2]. In these optofluidic devices, the total integration of microfluidic circuits and optical components offers great potential for multi-functionalities and miniaturization, which makes them suitable for chemical analysis in micrototal analysis system (μ TAS) and clinical diagnosis right at the point-of-care [3,4]. Up to now, sensing performance of the optofluidic sensors based on Fabry-Perot cavities [5–8], integrated interferometers [9–11], diffractive gratings [12–20], hollow waveguides [21–23], microring resonators [24–27], surface plasmon resonance [28–30], micro-structured optical fibers [31–35] and guided-mode resonance (GMR) [36–40] have been studied. The interferometer can determine the RI change of $\sim 10^{-7}$ RIU by measuring the spatial shift of the interference pattern [9]. A capillary-based optofluidic ring resonator with an outer radius of 35 μm and a wall thickness of ~ 1 μm typically has a detection limit of 10^{-7} RIU due to its high Q-factor over 10^5 [27]. The novel GMR device achieves a superior RI resolution of 10^{-5} RIU over a wide linear detection range according to the variation in the light intensity [37]. A step-index fiber with a parallel hollow micro-channel provides a high sensitivity of up to 3200 nm/RIU [35]. Diffractive gratings, as one of the typical diffractive microphotonic devices can detect $\sim 10^{-5}$ RI change [17,18,20], which have drawn a lot of attentions because of following advantages. Firstly, the measurement principles are always based on the readout of the diffraction patterns, which only requires inexpensive cameras instead of complicated facilities such as high-resolution spectrometers and bulky optics. Secondly, the planar devices are inherently feasible to be integrated into a lab-on-chip system without any additional structural design, sophisticated fabrication processes and special process treatments. Finally, the environmental noises can be effectively suppressed by using self-referencing differential sensing in a diffraction pattern, in which the signal of interest and the reference interfere in the same path.

In this paper, we introduce a novel optofluidic lab-on-chip device to detect the RI based on asymmetric single-slit Fraunhofer diffraction. The proposed device is operated by delivering half of a coherent laser beam to a slit via analyte and the other half directly to the slit, by which the two parts of the beam subsequently interfere and diffract from the slit. The detection mechanism relies on detecting the phase change of the optical wave induced by the RI change of the analyte. Thus, it is possible to measure small difference in the bulk RI, without the requirement that the RI of analyte have to be higher than the cladding to form waveguides or resonators. The device was fabricated in polydimethylsiloxane (PDMS) microchannel system using standard soft lithography technique as our previous work [41], which is facile and suitable for low-cost production. A laser beam illuminates the analyte in front of a slit and forms an asymmetric diffraction pattern behind based on the intensity modulation effect without interacting with the fluidic itself. An in-plane optofluidic lens was integrated to the microfluidic networks to avoid the manual alignment of the optical components as well as to reduce the cost of external bulky components [42,43]. In addition, a diffraction pattern visualization zone (or called a “screen”) filled with the dye solution of Rhodamine B was designed and placed in the chip to in-plane visualize the sensing fluorescence patterns. It is actually similar to a quick response code (QR code) which is readily read-out for clinical diagnosis right at the point-of-care or on-site security check. The new device is practical and suitable to be extended for high throughput applications by simultaneously reading multiple chips with an 2D-array image sensor. We utilize the device to achieve a low NEDL of $\sim 10^{-6}$ RIU and high sensitivity of close to 1.1×10^4 /RIU within a confidence level of 95% as verified by Student's t-test. This optofluidic device will provide a fully integrated, simple, inexpensive sensing platform for sensitive and fast RI measurement.

2. Device design and theory

A schematic illustration of the optofluidic chip is shown in Fig. 1. The chip consists of a pair of microchannels placed to each other, in which the black ink is injected to form a slit with an opening of $W = 330 \mu\text{m}$. A microfluidic passive micromixer with one inlet fed by de-ionized (DI) water and the other fed by a mixture of glycerin and DI water, in which mixed homogenous analyte (the length of sensing microchannel $L = 750 \mu\text{m}$) is located $L_1 = 700 \mu\text{m}$ away in front of the slit. The volume ratio of the water to glycerin can be adjusted using syringe pumps during the experiment through the micromixer, in order to change the RI without stopping the sensor and also preventing the formation of the bubbles inside the microchannels. A liquid-core liquid-cladding (L^2) optofluidic biconvex microlens is developed in a circuit chamber behind the slit. The diameter of the chamber is $D = 500 \mu\text{m}$ and the distance to the slit is $L_2 = 895 \mu\text{m}$. When the core liquid sandwiched by two cladding liquids enters the circular chamber, it follows the arc-shape streamlines inside the chamber and, therefore, forms the typical shape of a spherical biconvex microlens. In order to easily determine the sensing image, a diffraction pattern visualization zone, which can be taken as a screen, is designed and placed behind the microlens, where we injected fluorescent dye solution of Rhodamine B into this visualization zone. A single-mode optical fiber with a numerical aperture of $\text{NA} = 0.22$ was inserted into a pre-fabricated fiber groove with a width of $130 \mu\text{m}$ to work approximately as a point light source, which is about $L_3 = 3.85 \text{ mm}$ away from the sensing microchannel. All the components have a height of $150 \mu\text{m}$ and are designed to be located at the same altitude. When a 532 nm laser beam launches from the fiber, it propagates in the chip plane with half of its wavefront interacting with the analyte. Then the modulated light is diffracted by the slit and focused by a microlens, leading to a fluorescence diffraction pattern in the “screen”. The asymmetric distribution of the pattern reveals the phase change of the optical wave that is induced by the RI change of the analyte. This allows us to detect the RI by measuring the intensity ratio I_m/I_{-m} , where $I_{\pm m}$ is the $\pm m$ -order diffraction fringes.

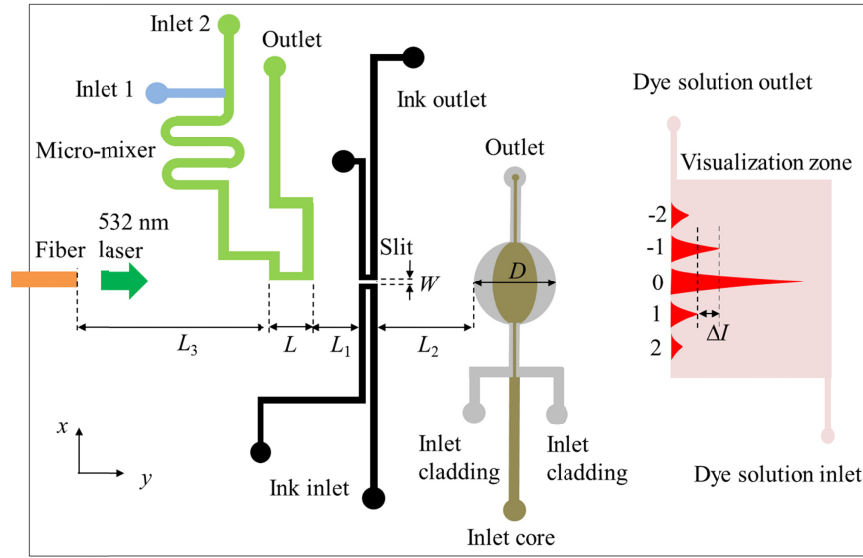


Fig. 1. A schematic illustration of the optofluidic chip. The chip consists of a inserted fiber, a slit, microchannel networks, micromixers, an in-plane optofluidic biconvex microlens and a visualization zone for displaying the fluorescence diffraction pattern.

The dependence of the intensity distribution of the diffraction pattern produced by the slit on the RI change Δn can be calculated based on the Fraunhofer diffraction theory [44,45]. Half of the laser beam traveled through the analyte in the microfluidic channel while the other half traveled through PDMS. This arrangement split the initial beam into two beams of almost equal intensity, neglecting the low optical absorption of the analyte and PDMS. The Fraunhofer diffraction complex fields of the two beams U_1 and U_2 are given by Fresnel-Kirchhoff diffraction integral on their own sides of the slit, respectively:

$$U_1 = -\frac{Ai \cos \delta}{\lambda rs} \int_{-a}^{\epsilon} e^{i[k(r+s)+\Delta\phi]} d\xi, \quad (1)$$

$$U_2 = -\frac{Ai \cos \delta}{\lambda rs} \int_{\epsilon}^a e^{ik(r+s)} d\xi, \quad (2)$$

where λ is the light wavelength; $k = 2\pi n_{\text{PDMS}}/\lambda$ is the wave vector; A is a constant account for the field amplitude; δ is the observation angle relative to the backward normal to the slit; r and s are the distance from the light source and the points in the “screen” to a particular point along slit, respectively; $2a$ is the width of the slit; ϵ is the boundary coordinate of the two beams; $\Delta\phi$ is phase difference between the two beams which can be written as:

$$\Delta\phi = k(n_{\text{analyte}} - n_{\text{PDMS}})L, \quad (3)$$

where n_{analyte} and n_{PDMS} are the refractive indices of the analyte and the PDMS. With the two beams recombining in the “screen”, the phase difference between them will cause interference in the diffraction pattern intensity. Hence, the intensity of the diffraction peak as a function of the position along the slit width can be described by

$$I = |U_1 + U_2|^2, \quad (4)$$

Substituting Eqs. (1)-(3) in Eq. (4), the spatial intensity distribution $I(x)$ along the direction (x axis) of the slit opening is given by

$$I(x) = C \frac{4s'^2}{k^2 x^2} \left[1 + 2 \sin \frac{\Delta\phi}{2} \cos \left(k \frac{x}{s'} \varepsilon \right) \sin \left(k \frac{x}{s'} a - \frac{\Delta\phi}{2} \right) - \cos \left(k \frac{x}{s'} a \right) \cos \left(k \frac{x}{s'} a - \Delta\phi \right) \right], \quad (5)$$

where C is a constant, s' is the distance from the slit center to the “screen”, and x is the coordinate along the direction of the slit opening. Note that Eq. (5) is a periodic function of the phase difference $\Delta\phi$ and the change of $\Delta\phi$ causes both an intensity change and a shift in the position of a diffraction fringe. Figure 2(a) shows the calculated diffraction pattern for varied RI change Δn of the analyte. Note that the optical intensity of the 1st diffraction orders grow up dramatically while the -1^{st} diffraction orders fall down with the increase of the RI change. Here we define a dimensionless parameter as the transducing signal for a pair of modes as

$$S = \ln(I_1 / I_{-1}), \quad (6)$$

in our measurement and data processing, similar to our previous work [46]. This parameter S reveals the intensity difference between the pair of modes, and the natural logarithm transformation is performed to achieve better linearity. Figure 2(b) shows the calculated S as a function of the RI change Δn of the analyte. This is a linear function of the RI change for a small detection range and the slope of curve indicates that the calculated sensitivity is about $1.3 \times 10^4/\text{RIU}$. The inset shows the transducing signal S as a function of the phase difference $\Delta\phi$ which indicates S is a periodic function (period = 2π). The curve in the red circle has a steep and linear slope so that it is used to achieve high RI sensitivity in our experiment.

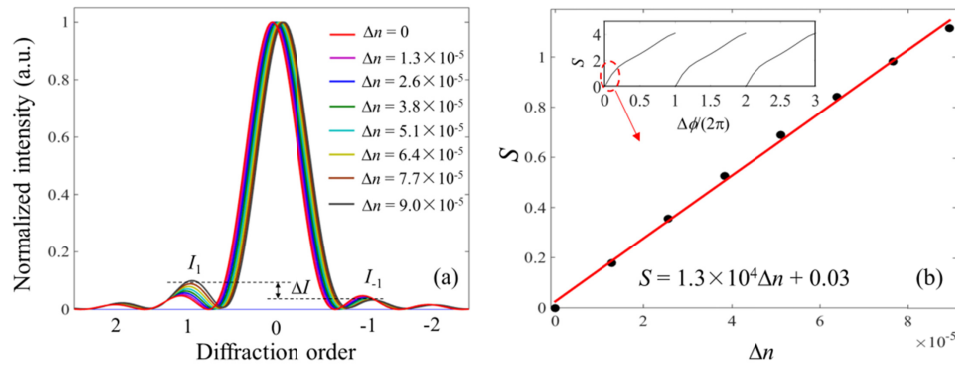


Fig. 2. (a) Calculated diffraction patterns for varied RI change Δn of the analyte. The optical intensity of the 1st diffraction orders grow up dramatically while the -1^{st} diffraction orders fall down with the increase of the RI change Δn . (b) Calculated transducing signal S as a function of the RI change Δn . The slope of the curve indicates that the calculated sensitivity is about $1.3 \times 10^4/\text{RIU}$. The inset shows the transducing signal S as a function of the phase difference $\Delta\phi$ which indicates S is a periodic function. The curve in the red circle has a steep and linear slope so that it is used to achieve high RI sensitivity in our experiment.

3. Fabrication and experimental setup

The device was fabricated in PDMS using standard soft lithography technique. The slit, micromixer, fiber groove, visualization zone and microlens including the microchannel network was printed on a transparency film with a resolution of 8000 dpi. The transparency mask was subsequently used for defining the negative mold of the lens in a 150 μm thick SU-8 layer. The PDMS was mixed in a weight ratio of 10:1 and poured onto the SU-8 mold as a master. Subsequently, the PDMS was placed in vacuum for 2 hours for degassing. After curing, the PDMS piece was then peeled off from the master mould, and bonded to another

flat PDMS part after treating both surfaces with oxygen plasma for 40 seconds. The bonded PDMS was then placed in an oven at 150 °C for 2 hours to ensure good bonding between both surfaces. A single-mode optical fiber was inserted into a pre-fabricated fiber groove to work approximately as a point light source. Finally, black ink was injected into a pair of microchannels to form a slit while fluorescence dye Rhodamine B (Sigma-Aldrich, excitation wavelength of 540 nm, emission wavelength of 625 nm) diluted in a mixture ($n = 1.412$) of glycerin (60% by weight) and DI water (40% by weight) was injected in to the visualization zone for diffraction patterns. The microlens (focal length = 2.07 mm) has a liquid core (cinnamaldehyde, $n = 1.62$) and a liquid cladding (a mixture of 73.5% ethylene glycol and 26.5% ethanol by weight, $n = 1.412$), at a flow rate of 60 $\mu\text{l}/\text{min}$ and 10 $\mu\text{l}/\text{min}$, respectively. Applying this technology, an optofluidic on-chip diffraction demonstration system, as well as the microfluidic network with height of 150 μm was realized in Fig. 3(a).

The experimental setup of the measurement is shown in Fig. 3(b). We used 5 syringe pumps (two for the inlets of the micromixer, two for the cladding and core of the microlens and one for the visualization zone) to inject the fluids in the optofluidic chip and the waste was collected into a glass cup. The input optical light was coupled into the optofluidic chip by a single-mode optical fiber connected to a semiconductor laser (Laserwave, LWGL 532-100 mW-F) through a coupling system of spatial light to optical fiber. The output fluorescence which was projected in the diffraction pattern visualization zone was collected by an objective and then sent to a CCD camera that recorded the intensity.

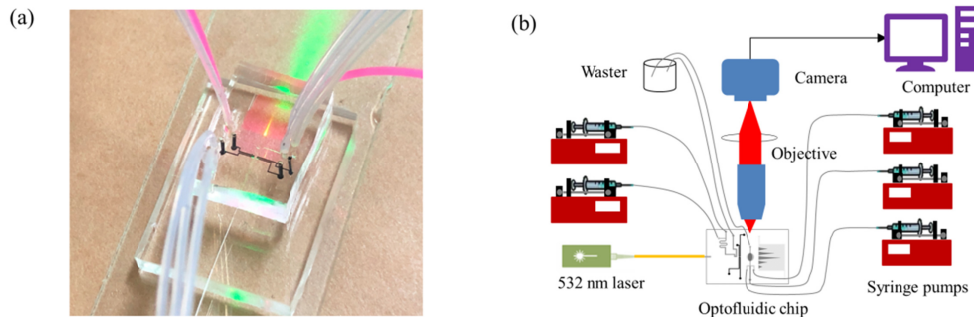


Fig. 3. (a) Optofluidic chip: Laser beam passes through the sensing microchannel, a slit and then a diffraction pattern is displayed in the visualization zone. (b) The experimental setup: syringe pumps, laser, objective, camera and computer.

The protocol of use of the sensor can be defined as the following steps: 1) Inject the reference liquid into the microchannel in front of the slit. Capture the image of the diffraction pattern; 2) Inject the analyte to be detected into the microchannel in front of the slit under the same experimental conditions. Capture a new image of the diffraction pattern; 3) Analyze the two images of diffraction patterns with in-house image recognition and algorithm, to locate the $\pm m$ -order diffraction fringes and calculate the transducing signals S ; 4) Predict the RI difference of the analyte from the reference liquid with the transducing signals S of both them and the known sensitivity of the device.

To characterize the bulk RI detection capability of the device, the analyte with varied homogenous refractive indices was prepared by using the conventional microfluidic technology which can generate homogenous analyte for biosensing experiments [47,48]. As shown in Fig. 4(a), a passive micromixer consisting of a serpentine channel with 12 windings was used to mix the high-RI liquid (a mixture of glycerin and DI water, $n_1 = 1.33416$, flow rate is Q_1) and the low-RI liquid (DI water, $n_2 = 1.33303$, flow rate is Q_2). The RI of the mixed liquid was changed by adjusting ratio of Q_2 to Q_1 . We have performed the computational fluid dynamics simulations based on finite element method, along with the mass balance equations and steady boundary conditions to show the mixing capability of the

micromixer. The simulations performed by choosing different sets of (Q_1 , Q_2) show uniform color at each outlet, which reveal that the liquids are well-mixed after passing the micromixer. The RI of the mixed liquid can be calculated as $n = (n_1Q_1 + n_2Q_2)/(Q_1 + Q_2)$ [49], and Fig. 4(b) presents the relation between the RI of the mixed liquid and Q_2 for a fixed Q_1 of 80 $\mu\text{l}/\text{min}$. This figure shows that the RI decreases linearly as the flow rate of the low-RI liquid (Q_2) increases. Since the typical syringe pump (KDS Model 200 Series) has the minimum flow rate of 5.746 $\mu\text{l}/\text{hr}$ [50] which is one order of magnitude lower than the minimum flow rate of 1 $\mu\text{l}/\text{min}$ in our experiment, the mixing ratio of two liquids is believed to be controlled accurately.

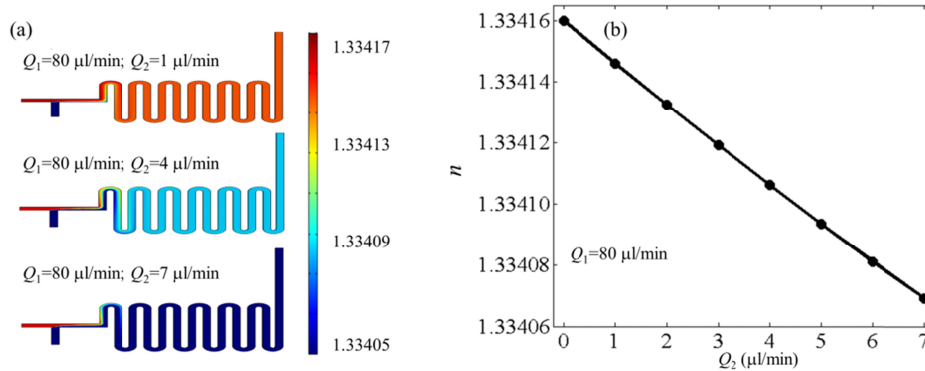


Fig. 4. (a) The simulation results for the mixing of DI water and glycerin in three different sets of flow rates (Q_1 , Q_2). The homogenous color at the out-ports reveal that the liquids are well-mixed (b) The relation between the RI of the mixed liquid and Q_2 for a fixed Q_1 of 80 $\mu\text{l}/\text{min}$. The RI decreases nearly linearly as the flow rate of the low-RI liquid (Q_2) increases.

4. Results and discussions

Figure 5(a) shows the 2D diffraction patterns before ($n = 1.33416$) and after ($n = 1.33409$) analyte RI change. We see that the diffraction patterns are not mirror-symmetric ($I_m \neq I_{-m}$) as the conventional Fraunhofer diffraction patterns when the RI of the analyte is different from that of PDMS. Importantly, the ratio of the intensity of the $\pm m$ -order diffraction fringes increases dramatically when the RI of the analyte decreases slightly while the zero-order diffraction fringe at the center remains almost unchanged. This physical phenomenon inherently leads to an ultrahigh RI detection sensitivity. The intensity of the diffraction fringes is extracted by integrating the intensity along the direction of the beam propagation (x-axis) in the images. In this way, the 2-D image is converted to a 1-D curve. As for the detection limit of the sensor, it is determined by not only the detection sensitivity but also the noise of the system. Thus it is critical to achieve a low-noise setup. The experiment was performed at 20 $^{\circ}\text{C}$ with a system temperature fluctuation was kept within 0.1 $^{\circ}\text{C}$ in order to eliminate the thermal effect during the measurement. Figure 5(b) shows the recoding of signal S every second without changing the RI of the sample ($n = 1.33416$) for 300 seconds. The standard deviation of the signal, σ_1 , is 0.014 due to the low-frequency fluctuation of the noises, which is higher than the typical performance in other intensity-based detection biosensor systems [36–40]. We believe that the main contribution of noise comes from the fluctuation of the analyte flow driven by syringe pumps which have a nut-screw mechanism with step-by-step injection. Other noise contributions could be subjected to the laser power instability, uncompensated thermal fluctuation, signal-to-noise ratio of the CCD and stray light of the environment. These noises will result in very small optical intensity change or the slight shift of the positions of pattern fringes, which cause the errors of transducing signal S . However, for the real application scenarios, the micromixer driven by syringe pumps will not

be used for preparing various-RI analytes. So the low noise (superior resolution) could be expected in real applications.

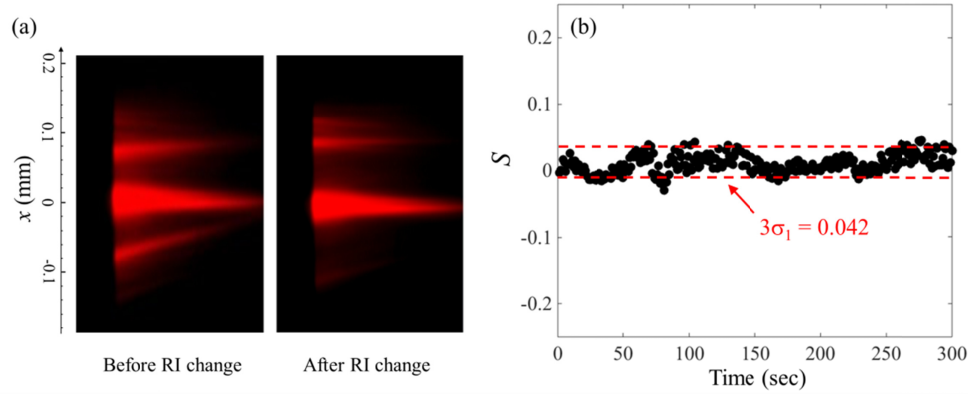


Fig. 5. (a) 2D diffraction patterns before ($n = 1.33416$) and after ($n = 1.33409$) analyte RI change. The diffraction patterns are asymmetric and the ratio of the intensity of the $\pm m$ -order diffraction fringes increases dramatically when the RI of the analyte decreases slightly. (b) The transducer signal S as a function of the measurement time without changing the RI of the analyte ($n = 1.33416$).

The bulk RI sensitivity of the sensor is characterized by monitoring the signal of measurement S in response to a change in the RI of the analyte upon the dynamic mixing of glycerin and DI water in the micromixer. This characterization indicates the RI sensing capability of the device. The flow rates (Q_2) of the DI water ($n = 1.33303$) were set to be from $1 \mu\text{l/min}$ to $7 \mu\text{l/min}$, respectively, with a constant flow rate (Q_1) of the glycerin-water mixture ($n = 1.33416$) at $80 \mu\text{l/min}$. Before injecting the DI water, a stable detection baseline of the system was first established to ensure the micromixer initially filled with glycerin-water mixture had reached thermal equilibrium. When the DI water was then introduced into the micromixer, the RI decrease of the analyte resulted in a change in the signal S . Figure 6(a) plots a ladder-like sensorgram, in which the signal S was tracked 100 times for every different low-RI flow rate Q_2 . The signal S is in proportion to the RI change of the analyte for a small detection range, which is marked as dark dots in Fig. 6(b). The red curve shows the linearly fitted curve, and the slope indicates that the experimental bulk RI sensitivity ($s = dS/dn$) is close to $1.1 \times 10^4/\text{RIU}$ within a confidence level of 95% as verified by Student's t-test. This shows a good agreement with the calculation results ($\sim 1.3 \times 10^4/\text{RIU}$) in Fig. 2(b). This difference mainly comes from the "impreciseness" of the circuit that are naturally present due to the chip fabrication tolerance and various system noise contributions which were analyzed previously. The standard deviation of the signal, σ_2 , is 0.041, so the current NEDL is estimated as $\text{NEDL}_2 = 3\sigma_2/s = 1.1 \times 10^{-5} \text{ RIU}$. Note that σ_2 is slightly greater than σ_1 which was derived from Fig. 5(b). This is probably because of large fluctuation of the analyte flow or larger system instability when changing the flow rates of the syringe pumps manually. This unwanted additional noise can be eliminated eventually when measuring real sample in no need of the syringe pumps to prepare various-RI analytes. Therefore the expected NEDL can be estimated as $\text{NEDL}_1 = 3\sigma_1/s = 3.8 \times 10^{-6} \text{ RIU}$. In addition, we tested tens of additional devices and achieved the high sensitivity in the order of magnitude of $\sim 10^4/\text{RIU}$, which show good reproducibility.

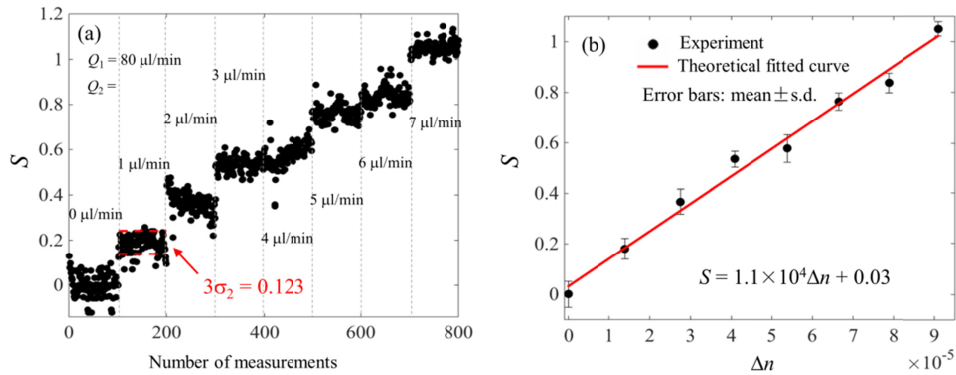


Fig. 6. (a) The ladder-like signal S sensorgram corresponding to the flow rate of DI water $Q_2 = 1\text{--}7\text{ }\mu\text{L}/\text{min}$. The signal S was tracked 100 times for every different low-RI flow rate Q_2 . (b) The experimental transducing signal S as a function of the RI change. The RI of the analyte is varied by changing the flow rate (Q_2) of DI wafer from 1 $\mu\text{L}/\text{min}$ to 7 $\mu\text{L}/\text{min}$ with a constant flow rate (Q_1) of the glycerin-water mixture at 80 $\mu\text{L}/\text{min}$. The slope of the theoretical fitted curve indicates that the sensitivity is close to $1.1 \times 10^4/\text{RIU}$ within a confidence level of 95% as verified by Student's t-test. Error bars: one standard deviation.

5. Conclusions

In summary, we proposed a novel optofluidic sensor chip for RI detection with ultra-small change. The sensor is based on asymmetric Fraunhofer diffraction, relying on detecting the phase change of the optical wave induced by the RI change of analyte. In the experiment, the device can measure a RI change of as low as $\sim 10^{-5}$ RIU. A low NEDL of $\sim 10^{-6}$ RIU and high sensitivity of close to $1.1 \times 10^4/\text{RIU}$ within a confidence level of 95% as verified by Student's t-test were achieved and agree well with the theory model. This optofluidic device may provide a fully integrated, simple, inexpensive sensing platform for sensitive and fast RI measurement.

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References

1. X. Fan, I. M. White, S. I. Shopova, H. Zhu, J. D. Suter, and Y. Sun, "Sensitive optical biosensors for unlabeled targets: a review," *Anal. Chim. Acta* **620**(1-2), 8–26 (2008).
2. D. Psaltis, S. R. Quake, and C. Yang, "Developing optofluidic technology through the fusion of microfluidics and optics," *Nature* **442**(7101), 381–386 (2006).
3. C. Song, N.-T. Nguyen, and S. H. Tan, "Toward the commercialization of optofluidics," *Microfluid. Nanofluidics* **21**(8), 139 (2017).
4. C. Song and H. S. Tan, "A perspective on the rise of optofluidics and the future," *Micromachines (Basel)* **8**(5), 152 (2017).
5. P. Domachuk, I. C. M. Littler, M. Cronin-Golomb, and B. J. Eggleton, "Compact resonant integrated microfluidic refractometer," *Appl. Phys. Lett.* **88**(9), 093513 (2006).
6. R. St-Gelais, J. Masson, and Y. A. Peter, "All-silicon integrated Fabry-Pérot cavity for volume refractive index measurement in microfluidic systems," *Appl. Phys. Lett.* **94**(24), 243905 (2009).
7. W. Z. Song, X. M. Zhang, A. Q. Liu, C. S. Lim, P. H. Yap, and H. M. M. Hosseini, "Refractive index measurement of single living cells using on-chip Fabry-Pérot cavity," *Appl. Phys. Lett.* **89**(20), 203901 (2006).
8. Y. Guo, H. Li, K. Reddy, H. S. Shelar, V. R. Nittoor, and X. Fan, "Optofluidic Fabry-Pérot cavity biosensor with integrated flow-through micro-/nanochannels," *Appl. Phys. Lett.* **98**(4), 041104 (2011).

9. A. Ymeti, J. S. Kanger, R. Wijn, P. V. Lambeck, and J. Greve, "Development of a multichannel integrated interferometer immunosensor," *Sens. Actuators B Chem.* **83**(1–3), 1–7 (2002).
10. A. Crespi, Y. Gu, B. Ngamsom, H. J. W. M. Hoekstra, C. Dongre, M. Pollnau, R. Ramponi, H. H. van den Vlekert, P. Watts, G. Cerullo, and R. Osellame, "Three-dimensional Mach-Zehnder interferometer in a microfluidic chip for spatially-resolved label-free detection," *Lab Chip* **10**(9), 1167–1173 (2010).
11. M. I. Lapsley, I. K. Chiang, Y. B. Zheng, X. Ding, X. Mao, and T. J. Huang, "A single-layer, planar, optofluidic Mach-Zehnder interferometer for label-free detection," *Lab Chip* **11**(10), 1795–1800 (2011).
12. C. Chang, Z. Ding, V. N. L. R. Patchigolla, B. Ziaie, and C. A. Savran, "Reflective diffraction gratings from hydrogels as biochemical sensors," *IEEE Sens. J.* **12**(7), 2374–2379 (2012).
13. O. J. A. Schueller, D. C. Duffy, J. A. Rogers, S. T. Brittain, and G. M. Whitesides, "Reconfigurable diffraction gratings based on elastomeric microfluidic devices," *Sens. Actuators A Phys.* **78**(2–3), 149–159 (1999).
14. C. Lv, Z. Jia, Y. Liu, J. Mo, P. Li, and X. Lv, "Angle-resolved diffraction grating biosensor based on porous silicon," *J. Appl. Phys.* **119**(9), 094502 (2016).
15. H. Yu, G. Zhou, F. S. Chau, and F. Lee, "Phase-transmission-grating-based compact optofluidic refractometer," *Opt. Lett.* **34**(12), 1753–1755 (2009).
16. S. Calixto, N. C. Bruce, and M. Rosete-Aguilar, "Diffraction grating-based sensing optofluidic device for measuring the refractive index of liquids," *Opt. Express* **24**(1), 180–190 (2016).
17. F. Purr, M. Bassu, R. D. Lowe, B. Thürmann, A. Dietzel, and T. P. Burg, "Asymmetric nanofluidic grating detector for differential refractive index measurement and biosensing," *Lab Chip* **17**(24), 4265–4272 (2017).
18. L. F. G. Dib and E. A. Barbosa, "Immersed diffraction grating refractometers of liquids," *Appl. Opt.* **55**(30), 8582–8588 (2016).
19. G. Bettella R, R. Zamboni, G. Pozza, A. Zaltron, C. Montevecchi, M. Pierno, G. Mistura, C. Sada, L. Gauthier-Manuel, and M. Chauvet, "LiNbO₃ integrated system for opto-microfluidic sensing," *Sens. Actuators B Chem.* **282**, 391–398 (2019).
20. J. Q. Yu, Y. Yang, A. Q. Liu, L. K. Chin, and X. M. Zhang, "Microfluidic droplet grating for reconfigurable optical diffraction," *Opt. Lett.* **35**(11), 1890–1892 (2010).
21. A. Llobera, R. Wilke, and S. Büttgenbach, "Optimization of poly(dimethylsiloxane) hollow prisms for optical sensing," *Lab Chip* **5**(5), 506–511 (2005).
22. A. Llobera, S. Demming, R. Wilke, and S. Büttgenbach, "Multiple internal reflection poly(dimethylsiloxane) systems for optical sensing," *Lab Chip* **7**(11), 1560–1566 (2007).
23. B. Ibarlucea, C. Fernández-Sánchez, S. Demming, S. Büttgenbach, and A. Llobera, "Selective functionalisation of PDMS-based photonic lab on a chip for biosensing," *Analyst (Lond.)* **136**(17), 3496–3502 (2011).
24. M. Li, X. Wu, L. Liu, X. Fan, and L. Xu, "Self-referencing optofluidic ring resonator sensor for highly sensitive biomolecular detection," *Anal. Chem.* **85**(19), 9328–9332 (2013).
25. H. Zhu, I. M. White, J. D. Suter, and X. Fan, "Phage-based label-free biomolecule detection in an opto-fluidic ring resonator," *Biosens. Bioelectron.* **24**(3), 461–466 (2008).
26. I. M. White, J. Gohring, Y. Sun, G. Yang, S. Lacey, and X. Fan, "Versatile waveguide-coupled optofluidic devices based on liquid core optical ring resonators," *Appl. Phys. Lett.* **91**(24), 241104 (2007).
27. H. Li and X. Fan, "Characterization of sensing capability of optofluidic ring resonator biosensors," *Appl. Phys. Lett.* **97**(1), 011105 (2010).
28. F. Yu, S. Tian, D. Yao, and W. Knoll, "Surface plasmon enhanced diffraction for label-free biosensing," *Anal. Chem.* **76**(13), 3530–3535 (2004).
29. C. Escobedo, A. G. Brolo, R. Gordon, and D. Sinton, "Optofluidic concentration: plasmonic nanostructure as concentrator and sensor," *Nano Lett.* **12**(3), 1592–1596 (2012).
30. Z. Xiao, H. Dai, and X. Chen, "Ultrasensitive optofluidic resonator refractive index sensor," *Opt. Lett.* **43**(17), 4216–4219 (2018).
31. A. P. Zhang, G. Yan, S. Gao, S. He, B. Kim, J. Im, and Y. Chung, "Microfluidic refractive-index sensors based on small-hole microstructured optical fiber Bragg gratings," *Appl. Phys. Lett.* **98**(22), 221109 (2011).
32. A. M. Cubillas, S. Unterkofler, T. G. Euser, B. J. M. Etzold, A. C. Jones, P. J. Sadler, P. Wasserscheid, and P. S. J. Russell, "Photonic crystal fibres for chemical sensing and photochemistry," *Chem. Soc. Rev.* **42**(22), 8629–8648 (2013).
33. Z. He, F. Tian, Y. Zhu, N. Lavlinskaia, and H. Du, "Long-period gratings in photonic crystal fiber as an optofluidic label-free biosensor," *Biosens. Bioelectron.* **26**(12), 4774–4778 (2011).
34. C. Wu, M.-L. V. Tse, Z. Liu, B.-O. Guan, A. P. Zhang, C. Lu, and H.-Y. Tam, "In-line microfluidic integration of photonic crystal fibres as a highly sensitive refractometer," *Analyst (Lond.)* **139**(21), 5422–5429 (2014).
35. H. W. Lee, M. A. Schmidt, P. Uebel, H. Tyagi, N. Y. Joly, M. Scharer, and P. S. Russell, "Optofluidic refractive-index sensor in step-index fiber with parallel hollow micro-channel," *Opt. Express* **19**(9), 8200–8207 (2011).
36. Y. F. Ku, H.-Y. Li, W.-H. Hsieh, L.-K. Chau, and G.-E. Chang, "Enhanced sensitivity in injection-molded guided-mode-resonance sensors via low-index cavity layers," *Opt. Express* **23**(11), 14850–14859 (2015).
37. Y.-C. Lin, W.-H. Hsieh, L.-K. Chau, and G.-E. Chang, "Intensity-detection-based guided-mode-resonance optofluidic biosensing system for rapid, low-cost, label-free detection," *Sens. Actuators B Chem.* **250**, 659–666 (2017).
38. Y. Nazirizadeh, U. Bog, S. Sekula, T. Mappes, U. Lemmer, and M. Gerken, "Low-cost label-free biosensors using photonic crystals embedded between crossed polarizers," *Opt. Express* **18**(18), 19120–19128 (2010).

39. S.-F. Lin, F.-C. Chang, Z.-H. Chen, C.-M. Wang, T.-H. Yang, W.-Y. Chen, and J.-Y. Chang, "A polarization control system for intensity-resolved guided mode resonance sensors," *Sensors (Basel)* **14**(3), 5198–5206 (2014).
40. J. Adrian, S. Pasche, J.-M. Diserens, F. Sánchez-Baeza, H. Gao, M. P. Marco, and G. Voirin, "Waveguide interrogated optical immunosensor (WIOS) for detection of sulfonamide antibiotics in milk," *Biosens. Bioelectron.* **24**(11), 3340–3346 (2009).
41. C. Song, T. Jin, R. Yan, W. Qi, T. Huang, H. Ding, S. H. Tan, N.-T. Nguyen, and L. Xi, "Opto-acousto-fluidic microscopy for three-dimensional label-free detection of droplets and cells in microchannels," *Lab Chip* **18**(9), 1292–1297 (2018).
42. C. Song, N.-T. Nguyen, S.-H. Tan, and A. K. Asundi, "A tuneable micro-optofluidic biconvex lens with mathematically predictable focal length," *Microfluid. Nanofluidics* **9**(4-5), 889–896 (2010).
43. C. Song, N.-T. Nguyen, S.-H. Tan, and A. K. Asundi, "Modelling and optimization of micro optofluidic lenses," *Lab Chip* **9**(9), 1178–1184 (2009).
44. M. Born and E. Wolf, *Principles of Optics: Electromagnetic Theory of Propagation, Interference and Diffraction of Light*, 7th ed (Cambridge University, 1999).
45. A. Lipson, S. G. Lipson, and H. Lipson, *Optical Physics*, 4th ed (Cambridge University, 2011).
46. H. Li, L. Shang, X. Tu, L. Liu, and L. Xu, "Coupling variation induced ultrasensitive label-free biosensing by using single mode coupled microcavity laser," *J. Am. Chem. Soc.* **131**(46), 16612–16613 (2009).
47. X. Chen, T. Li, and Z. Hu, "A novel research on serpentine microchannels of passive micromixers," *Microsyst. Technol.* **23**(7), 2649–2656 (2017).
48. J. M. Zhu, Y. Shi, X. Q. Zhu, Y. Yang, F. H. Jiang, C. J. Sun, W. H. Zhao, and X. T. Han, "Optofluidic marine phosphate detection with enhanced absorption using a Fabry-Pérot resonator," *Lab Chip* **17**(23), 4025–4030 (2017).
49. M. Oraie and H. Latifi, "Real-time refractive index sensing by using liquid core/liquid cladding optofluidic waveguide," *Opt. Laser Technol.* **111**, 303–306 (2019).
50. <https://www.kdscientific.com/kds-270-270p-legacy-syringe-pump.html>